Gait IK of quadruped robot

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Chapter 1

Trot motion

1.1 Locomotion Command Representation

1.1.1 Desired Body Motion

The controller is fundamentally a velocity-tracking controller.The operator specifies a desired body motion represented by

$$u = \begin{bmatrix} v_x \\ v_y \\ \dot{\psi} \end{bmatrix}$$

where

- v_x is the desired forward velocity,
- v_y is the desired lateral velocity,
- $\dot{\psi}$ is the desired yaw rate,

1.2 Gait Scheduling

1.2.1 Contact Phase Matrix for trot motion

$$C = \begin{array}{c|cccc} & \phi = \frac{1}{4} & \phi = \frac{1}{2} & \phi = \frac{3}{4} & \phi = 1 \\ \hline FR & 1 & 1 & 1 & 0 \\ FL & 1 & 0 & 1 & 1 \\ RR & 1 & 1 & 1 & 0 \\ RL & 1 & 0 & 1 & 1 \end{array}$$

1.2.2 Phase Timing

Define:

- Stance Time
- Swing Time

1.3 Stance Controller Based on COM Motion

1.3.1 Commanded Body Motion

The desired locomotion command is

$$u = \begin{bmatrix} v_x \\ v_y \\ \omega \end{bmatrix}$$

where

- v_x is the forward velocity,
- v_y is the lateral velocity,
- ω is the body yaw rate.

During one stance interval of duration

$$T_{stance}$$

the desired body displacement is

$$\Delta x_{COM} = v_x T_{stance}$$

$$\Delta y_{COM} = v_y T_{stance}$$

and the desired body rotation is

$$\theta = \omega T_{stance}.$$

These quantities define the desired rigid-body motion of the robot during the stance phase.

1.3.2 Control Algorithm

At every stance update:

1. Receive command

$$(v_x, v_y, \omega)$$

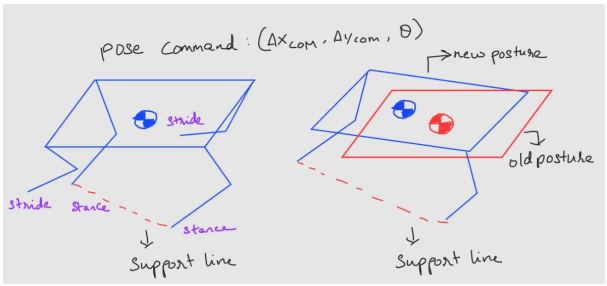
2. Compute the desired body motion during the stance interval

$$\Delta x_{COM} = v_x T_{stance}$$

$$\Delta y_{COM} = v_y T_{stance}$$

$$\theta = \omega T_{stance}.$$

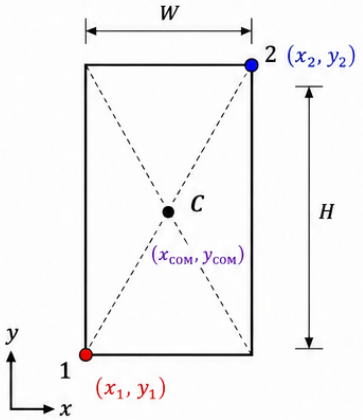
3. Compute the required hip joint displacements for the two diagonally opposite stance legs using the rigid-body equations shown in Figure 1.1b.



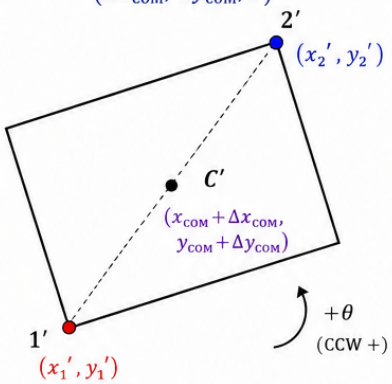
(a) Body motion generated during stance.

FROM COM COMMAND $(\Delta x_{com}, \Delta y_{com}, \theta)$ TO TRANSLATIONS OF DIAGONAL ENDS (x_1, x_2, y_1, y_2)

INITIAL CONFIGURATION



AFTER APPLYING COMMAND $(\Delta x_{com}, \Delta y_{com}, \theta)$



GIVEN COMMAND

- COM translation:
  - Δx_{com} (along $+x$)
  - Δy_{com} (along $+y$)
- Rotation:
  - θ (CCW positive)

FIND

(x_1', y_1', x_2', y_2')

KEY IDEA

Move the center by $(\Delta x_{com}, \Delta y_{com})$ and rotate the body by θ about the new center.
The diagonals keep the same length and remain straight after rigid motion.

1) INITIAL DIAGONAL GEOMETRY (IN BODY FRAME)

Vectors from COM to the diagonal ends in the initial (body) frame:

$$\boldsymbol{r}_1 = \begin{bmatrix} -\frac{W}{2} \\ -\frac{H}{2} \end{bmatrix} \quad \text{(to end 1)}$$

$$\boldsymbol{r}_2 = \begin{bmatrix} \frac{W}{2} \\ \frac{H}{2} \end{bmatrix} \quad \text{(to end 2)}$$

These vectors are fixed in the body and only rotate with the body.

2) ROTATION MATRIX

For rotation θ (CCW positive):

$$\boldsymbol{R}(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

3) NEW COM POSITION

New center of mass after translation:

$$\boldsymbol{C}' = \begin{bmatrix} x_{com} + \Delta x_{com} \\ y_{com} + \Delta y_{com} \end{bmatrix}$$

After rotation, the vectors to the ends become:

$$\boldsymbol{r}_1' = \boldsymbol{R}(\theta)\boldsymbol{r}_1 \quad , \quad \boldsymbol{r}_2' = \boldsymbol{R}(\theta)\boldsymbol{r}_2$$

4) POSITIONS OF THE DIAGONAL ENDS AFTER MOTION

End 1 (bottom-left diagonal):

$$\begin{bmatrix} x_1' \\ y_1' \end{bmatrix} = \boldsymbol{C}' + \boldsymbol{R}(\theta)\boldsymbol{r}_1$$

End 2 (top-right diagonal):

$$\begin{bmatrix} x_2' \\ y_2' \end{bmatrix} = \boldsymbol{C}' + \boldsymbol{R}(\theta)\boldsymbol{r}_2$$

$$\begin{aligned} x_1' &= (x_{com} + \Delta x_{com}) - \frac{W}{2}\cos\theta + \frac{H}{2}\sin\theta \\ y_1' &= (y_{com} + \Delta y_{com}) - \frac{W}{2}\sin\theta - \frac{H}{2}\cos\theta \\ x_2' &= (x_{com} + \Delta x_{com}) + \frac{W}{2}\cos\theta - \frac{H}{2}\sin\theta \\ y_2' &= (y_{com} + \Delta y_{com}) + \frac{W}{2}\sin\theta + \frac{H}{2}\cos\theta \end{aligned}$$

FINAL EQUATIONS (FROM COMMAND TO END TRANSLATIONS)

Given command: $\Delta x_{com}, \Delta y_{com}, \theta$

Compute (x_1', y_1', x_2', y_2')

$$\begin{aligned} x_1' &= (x_{com} + \Delta x_{com}) - \frac{W}{2}\cos\theta + \frac{H}{2}\sin\theta \\ y_1' &= (y_{com} + \Delta y_{com}) - \frac{W}{2}\sin\theta - \frac{H}{2}\cos\theta \\ x_2' &= (x_{com} + \Delta x_{com}) + \frac{W}{2}\cos\theta - \frac{H}{2}\sin\theta \\ y_2' &= (y_{com} + \Delta y_{com}) + \frac{W}{2}\sin\theta + \frac{H}{2}\cos\theta \end{aligned}$$

Notes

- θ positive is CCW.
- W, H are width and height of the rectangle.
- Equations are exact for any translation and rotation.

(b) Corresponding hip displacement computation.

4. Obtain

$$(\Delta x_1, \Delta y_1, \Delta x_2, \Delta y_2).$$

5. Compute the desired hip positions

$$\mathbf{P}'_{H_i} = R(\theta)\,\mathbf{P}_{H_i} + \begin{bmatrix} \Delta x_{COM} \\ \Delta y_{COM} \end{bmatrix}.$$

6. Since the stance feet remain fixed on the terrain,

$$\mathbf{P}'_{F_i} = \mathbf{P}_{F_i}.$$

7. Compute the desired foot position relative to the new hip frame

$$\mathbf{P}_{F_i REL} = \mathbf{P}_{F_i} - \mathbf{P}'_{H_i}.$$

   This represents the location of the fixed foot with respect to the desired hip position.

8. Use the standard 3-DOF leg inverse kinematics solver to compute the required joint angles

$$(q_{abd}, q_{hip}, q_{knee}) = IK\left({}^{H_i}\mathbf{p}_{F_i}\right).$$

9. Send the computed joint commands to the leg actuators.

1.4 Swing Phase Controller

1.4.1 Role of the Swing Phase

During the swing phase, the foot is no longer in contact with the terrain and therefore cannot contribute to support or propulsion.

The objective of the swing controller is to reposition the swing foot such that an appropriate support polygon is established for the next stance phase.

Since the stance controller generates body motion through the commanded

$$(\Delta x_{COM}, \Delta y_{COM}, \theta),$$

the swing controller must place the foot so that the next support configuration is compatible with this new body pose.

1.4.2 Touchdown Location Generation

Let

$$p_{nom}$$

denote the nominal touchdown location of a swing foot expressed in the body frame.

For example,

$$p_{FR,nom} = \begin{bmatrix} \frac{L}{2} \\ -\frac{W}{2} \\ -h \end{bmatrix}$$

for the front-right leg and

$$p_{RL,nom} = \begin{bmatrix} -\frac{L}{2} \\ \frac{W}{2} \\ -h \end{bmatrix}$$

for the rear-left leg.

These nominal positions define the desired support geometry of the quadruped when standing.

During the stance phase, the body undergoes the commanded rigid-body motion

$$(\Delta x_{COM}, \Delta y_{COM}, \theta).$$

Since the stance feet remain fixed on the terrain, their coordinates expressed in the body frame change according to the inverse body motion

$$p_{stance,new} = R_z(-\theta)\left(p_{stance,old} - \begin{bmatrix} \Delta x_{COM} \\ \Delta y_{COM} \\ 0 \end{bmatrix}\right).$$

Thus, from the perspective of the moving body, the stance feet appear to move backward.

To restore the nominal support geometry at the next touchdown event, the swing foot must be placed ahead by the same amount.

The desired touchdown position is therefore obtained by applying the body motion to the nominal foot location

$$p_{TD} = R_z(\theta)p_{nom} + \begin{bmatrix} \Delta x_{COM} \\ \Delta y_{COM} \\ 0 \end{bmatrix}.$$

P_{TD} = touchdown location.

This placement ensures that after touchdown the new support pair occupies the same relative configuration with respect to the body as the previous stance pair occupied before the body motion occurred.

Consequently, the swing foot advances by exactly the amount that the body moved relative to the fixed stance feet during the preceding stance phase.

Use a smooth transition from P_{nom} to P_{TD} for natural motion.

1.5 Inverse Kinematics

1.5.1 Closed-Form Inverse Kinematics Solution

The hip abduction/adduction angle is

$$\theta_1 = \tan^{-1}\left(\frac{y}{x}\right) + \tan^{-1}\left(\frac{\sqrt{x^2 + y^2 - l_2^2}}{\pm l_2}\right)$$

where the sign depends on whether the leg belongs to the left or right side of the robot.

The knee angle is

$$\theta_3 = -\cos^{-1}\left(\frac{\left(\sqrt{\left(\sqrt{x^2 + y^2 - l_2^2} - l_1\right)^2 + z^2}\right)^2 - l_3^2 - l_4^2}{2l_3l_4}\right)$$

The hip flexion/extension angle is

$$\theta_2 = \tan^{-1}\left(\frac{z}{\sqrt{x^2 + y^2 - l_2^2} - l_1}\right) - \tan^{-1}\left(\frac{l_4 \sin(\theta_3)}{l_3 + l_4 \cos(\theta_3)}\right)$$

Thus, the complete inverse kinematics solution becomes

$$q = \begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \end{bmatrix}.$$

1.5.2 Auxiliary Variables

For compactness, the following auxiliary variables are introduced:

$$F = \sqrt{x^2 + y^2 - l_2^2}$$

$$G = F - l_1$$

$$H = \sqrt{G^2 + z^2}$$

$$D = \frac{H^2 - l_3^2 - l_4^2}{2 l_3 l_4}$$

Using these definitions, the inverse kinematics equations can be written more compactly as

$$\theta_1 = \tan^{-1}\left(\frac{y}{x}\right) + \tan^{-1}\left(\frac{F}{\pm l_2}\right)$$

$$\theta_3 = -\cos^{-1}(D)$$

$$\theta_2 = \tan^{-1}\left(\frac{z}{G}\right) - \tan^{-1}\left(\frac{l_4 \sin(\theta_3)}{l_3 + l_4 \cos(\theta_3)}\right)$$